MIL-C-5778C
22 June 1973
SUPERSEDING
MIL-C-5778B
1 December 1958

MILITARY SPECIFICATION

COVERS, AIRCRAFT COMPONENTS, GENERAL REQUIREMENTS FOR

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers general requirements for covers intended to protect the following aircraft parts:

Cockpit	Fuselage
Engine	Armament
Propeller	Miscellaneous.

1.2 Classification. Covers shall be of the following types, as specified (see 6.2):

Type I - Waterproof fabric (3.3.1.1)

Type II - Coated cloth (3.3.1.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

T-T-881	Twine, Cotton, Seine
V-F-106	Fastener, Slide, Interlocking
V-T-276	Thread, Cotton
CCC-C-419	Cloth, Duck, Cotton, Unbleached, Plied-Yarns (Army and Numbered)
CCC-D-950	Dyeing and Aftertreating Processes for Cotton Cloths
PPP-B-601	Box, Wood, Cleated Plywood
PPP-B-636	Box, Fiberboard

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Military

MIL-P-116	Preservation, Methods of
MIL-W-5665	Webbing, Textile, Cotton Warp
MIL-C-7642	Cloth, Coated, Cotton, Twill Weave, One Side Vinyl Resin Coated
MIL-C-10799	Cloth, Coated Cotton, Vinyl Coated, Fire and Mildew Resistant
MIL-R-17343	Rope, Nylon
MIL-C-19524	Cloth, Coated (for Aircraft Protectors)
MIL-C-20696	Cloth, Coated, Nylon, Waterproof
MIL-F-21840	Fastener Tape, Hook and Pile, Nylon
MIL-M-46032	Mildew-Resistant Treatment (for Fabrics) Copper Processes

STANDARDS

Federal

FED-STD-751	Stitch, Seam and Stitching

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-831	Test Reports, Preparation of
MS20230	Grommet, Metallic, Plain and Spur, With Washer, Type I and Type III
MS51700	Streamer, Warning

PUBLICATION

Air Force-Navy Aeronautical Bulletin

157	Colors, List of Standard Aircraft Camouflage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 General. In addition to the requirements herein, the requirements specified on the drawings for covers for the particular model of aircraft shall also apply.

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3.3 Materials. Materials shall be as specified herein. Where materials are not specifically designated, they shall be entirely suitable for the purpose.

3.3.1 Fabric

3.3.1.1 Type I

3.3.1.1.1 Unless otherwise specified, material conforming to MIL-C-7642 shall be used in the covers for transparent plastic canopies such as the bombardier and pilot compartments, nose section, turrets, sight blisters, et cetera. Material for engine covers shall conform to MIL-C-10799, type II, class 4.

3.3.1.1.2 For all other covers, unless otherwise specified, the material used shall be type III, 12.29 ounces per square yard minimum conforming to CCC-C-419. The material shall be finished in accordance with CCC-D-950, type I, class B. Warning streamers, when specified (see 6.2), shall be in accordance with MS51700-12. The streamers shall be long enough to permit a 12-inch length exposure when the cover is installed on an airplane.

3.3.1.2 Type II. Unless otherwise specified (see 6.2), the material for type II covers shall conform to MIL-C-19524 or MIL-C-20696, type I, class 2. Warning streamers shall be the same as for type I.

3.3.2 Thread. Unless otherwise specified, all thread used for sewing shall be in accordance with V-T-276, ticket No. 16, 4 ply, type 1B3, and shall be the same color as the cover material.

3.3.3 Metal fittings. Grommets, eyelets, and fasteners shall conform to applicable service standards.

3.3.4 Webbing (type I). Unless otherwise specified, the webbing used for attaching the covers and for all binding and reinforcement shall conform to MIL-W-5665, and shall be the same color as the cover material.

3.3.5 Tie cord. Unless otherwise specified, the tie cord used for attaching covers and holding them in place shall be 1/4-inch rope conforming to MIL-R-17343.

3.3.6 Twine. Unless otherwise specified, the twine used for wrapping ends of the tie cords shall conform to T-T-881.

3.3.7 Mildew proofing. The cotton thread, webbing, and ropes used in engine covers shall be treated in accordance with MIL-M-46032, type I organic solvent dispersion method, except the permissible weight increase shall be 15 percent and the metallic copper content shall be 0.2 to 0.4 percent by weight.

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3.3.8 Interlocking fasteners (type I). Except as otherwise specified herein, interlocking slide fasteners shall conform to V-F-106, type IV, style No. 1 or 3, class B, size H, wire-stirrup pull, metal-zinc alloy with natural finish, color 613 of Bulletin 157. An 18-inch fabric thong shall be looped through the wire-stirrup pull and shall be capable of withstanding a tension pull of 100 pounds.

3.4 Design and construction. The covers shall be so designed that when initially installed they will fit neatly and completely cover the part of the airplane for which they are intended. The edges of the covers shall fit snugly to prevent infiltration of driving rain, snow, or dust. The covers shall not contain any wrinkles or lines of stress concentration. The method of padding valleys on the edges of the covers shall be in accordance with figure 1.

3.4.1 The covers, when installed, shall be adjustable to and capable of withstanding a stretch loading of not less than 2 pounds per inch of width in all directions.

3.4.2 Seams and stitching. The seams and stitching used in the manufacture of covers shall be in accordance with FED-STD-751.

3.4.2.1 Type I. All sewing shall be done with stitch type 301 to form a type LSc-2 or equivalent seam. There shall be not less than eight nor more than 10 stitches per inch.

3.4.2.2 Type II. All sewing shall be done with stitch type 401 to form a type LSa-2 seam with 3/4 inch gage. There shall be not less than four nor more than six stitches per inch.

3.4.3 Hems. Covers shall have a hem 1-1/2 inches wide on all free edges. Each edge of each hem shall one row of stitching.

3.4.4 Slits. Provisions shall be made for obstructions such as struts, stay wires, shafts, or other airplane parts, by means of slits. Slits shall extend to the nearest free edge, unless another position is specified. Unless otherwise specified, the edges of each slit shall be closed by means of an interlocking fastener installed as shown on figure 2. Means for attaching covers and holding them in place shall be accomplished by means of ties, straps, ropes or fasteners on the edge to permit fastening them around fuselage, wings, or some adjacent fittings. The hardware or fittings used for securing the cover in place shall be padded and the padding permanently attached by sewing around the periphery to prevent contact with the aircraft skin.

3.4.5 Patches. The covers shall be provided with reinforcement patches where they come in contact with sharp corners or projections.

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3.4.6 Engine covers. The engine cover shall extend beyond the fire wall and the propeller hub. Typical designs are shown on figures 3 through 7.

3.4.6.1 Fasteners. Engine covers around the nacelles shall be closed with interlocking fasteners as specified herein. The interlocking fastener shall be offset 5° from the vertical centerline of the engine.

3.4.6.2 Sleeves for hot air duct. Two sleeves, each approximately 8 inches long and 13 inches in diameter, shall be attached to each engine cover to serve as a hot air duct when the free end of the sleeve is attached to a heater. One sleeve shall be so located as to coincide with the opening into the accessories compartment of the engine. The other sleeve shall be located at the nose section of the cowling. A piece of type I lightweight webbing conforming to MIL-W-5665, approximately 52 inches long and 1/2 inch wide, shall be sewn at the midpoint to each sleeve near its open end. The color of the webbing shall approximately match the color of the cover.

3.4.6.3 Sleeve for propeller blade. A sleeve approximately 10 inches long and of sufficient diameter to fit the propeller blade shall be attached to the engine cover for each propeller blade.

3.4.6.4 Grommets. Plain grommets conforming to MS20230 shall be placed on the underside of the cover to permit excess oil drainage.

3.4.7 Propeller and rotor blade covers. The propeller and rotor blade covers shall match the respective engine cover propeller sleeves or rotor head cover blade sleeves. A typical design is shown on figure 8.

3.4.8 Fuselage and armament covers. These covers shall be for the bombardier and pilot compartments, turrets, sight blisters, and all other gun openings.

3.4.9 Miscellaneous covers. Covers for ventilation duct openings, tail and nose wheel openings, landing gear wells, and any opening too large to be covered with 2-inch masking tape shall be provided, unless justification by the weapon system manufacturer can be substantiated that the covers are not required.

3.4.10 Snap fasteners (type II). If it is necessary to secure junctions between cover components, snap fasteners as shown on figure 9 shall be used.

3.4.11 Fastening tapes. Fastening tapes conforming to MIL-F-21840, type II, may be used for fastening or securing flaps, pockets, folds, et cetera, when required. The tapes shall be of the size and class to match the fabric material being used in the protective cover.

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3.5 Carrying case(s) (type II). When specified (see 6.2), a carrying case(s) shall be provided for storing the covers. The design of the carrying case(s) shall receive prior approval by the procuring activity. The case(s) shall be made from the same materials and construction features as used in fabricating type II covers. The design of the case shall include suitable means for closing. Metallic parts shall be corrosion resistant or plated to resist corrosion.

3.6 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.7 Identification of product. Each cover shall be marked for identification in accordance with MIL-STD-130, plus any other special marking required by the contract or purchase order.

3.8 Workmanship. The cover, including all parts and accessories, shall be uniform in quality and shall be free from irregularities which would adversely affect the appearance or serviceability of the cover.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the covers shall be classified as follows:

a. Preproduction testing . . . . . See 4.3

b. Quality conformance tests . . . . . See 4.4.

4.3 Preproduction testing (see 6.2)

4.3.1 Test sample. One cover shall be subjected to the tests specified in 4.3.3.

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4.3.2 Test report. After completion of the preproduction tests, a test report shall be prepared in accordance with MIL-STD-831 and three complete copies of the report furnished to the procuring activity.

4.3.3 Preproduction tests. The preproduction tests shall consist of all tests specified under 4.5.

4.4 Quality conformance tests. The quality conformance tests shall consist of the individual test.

4.4.1 Individual test. Each cover shall be subjected to an examination of product test as described under 4.5.

4.5 Test methods

4.5.1 Examination of product. The cover shall be inspected to determine compliance with the requirements specified herein and on the applicable drawings with respect to materials, workmanship, and dimensions.

4.5.2 Installation. The cover shall be installed on the aircraft component being covered and examined for fit and alignment. It shall be demonstrated that the cover will withstand normal usage and handling encountered during installation, removal, and storage.

4.6 Inspection of preparation for delivery. Preservation, packaging, packing, and marking shall be inspected to determine conformance to the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be the same for both levels A and C.

5.1.1 Cleaning. Cleaning shall be in accordance with process C-1 of MIL-P-116.

5.1.2 Unit packaging. The covers shall be prepared for delivery in accordance with MIL-STD-794. The method of packaging shall conform to method III of MIL-P-116.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The covers, packaged as described in 5.1, shall be packed in a wood-cleated plywood container conforming to PPP-B-601, oversease type, style optional. When weight does not exceed 70 pounds, the covers may be packed in a container conforming to PPP-B-636, weather resistant. Closure and strapping shall be in accordance with the applicable container specification.

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5.2.2 Level B. The covers, packaged as described in 5.1, shall be packed in the wood-cleated plywood container conforming to PPP-B-601, domestic type, style optional. When weight does not exceed 70 pounds, the covers may be packed in a container conforming to PPP-B-636, domestic.

5.2.3 Level C. The covers, packaged as described in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.2 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. The following additional information shall be included:

Cover, Aircraft (Cockpit-Engine-Propeller-Fuselage-Armament-Miscellaneous, as applicable)

Airplane Model Designation

Drawing No. _____ (USAF or Navy, as applicable).

6. NOTES

6.1 Intended use. Covers conforming to this specification are intended to protect aircraft components from the weathering elements.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Quantity and type of cover desired (see 1.2)

c. Material for type II covers if other than specified (see 3.3.1.2)

d. Type of aircraft for which the cover will be used

e. Whether warning streamers are required (see 3.3.1.1.2)

f. When carrying cases are required (see 3.5)

g. Location and conditions for preproduction testing (see 4.3)

h. Applicable levels of packaging and packing (see 5.1 and 5.2).

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6.3 Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Air Force - 11
Army - AV
Navy - AS

Reviewers:
Air Force - 82
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Navy - AS

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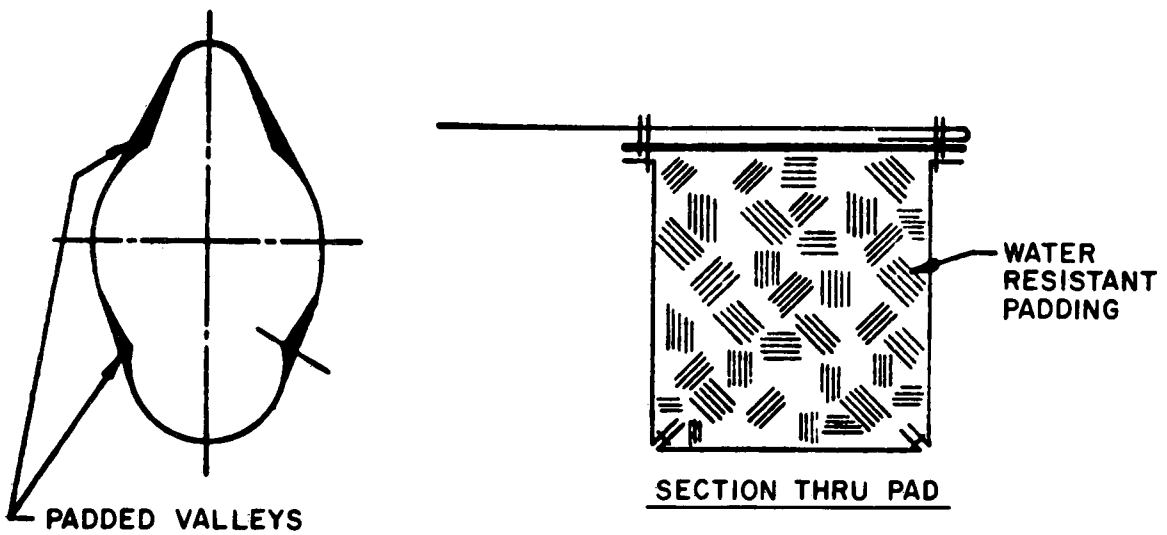


FIGURE 1. Method of Padding Valleys on Edge of Engine and Cockpit Covers.

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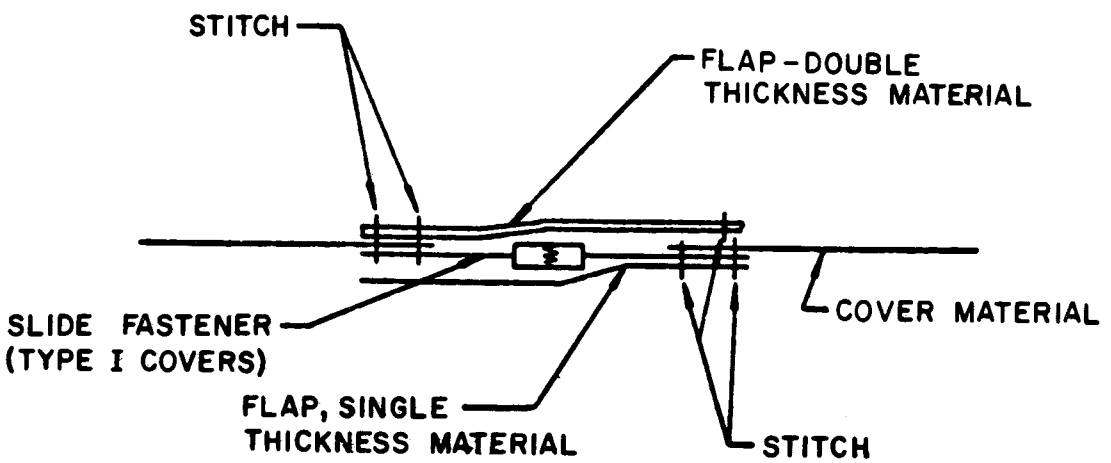


FIGURE 2. Cover Closure

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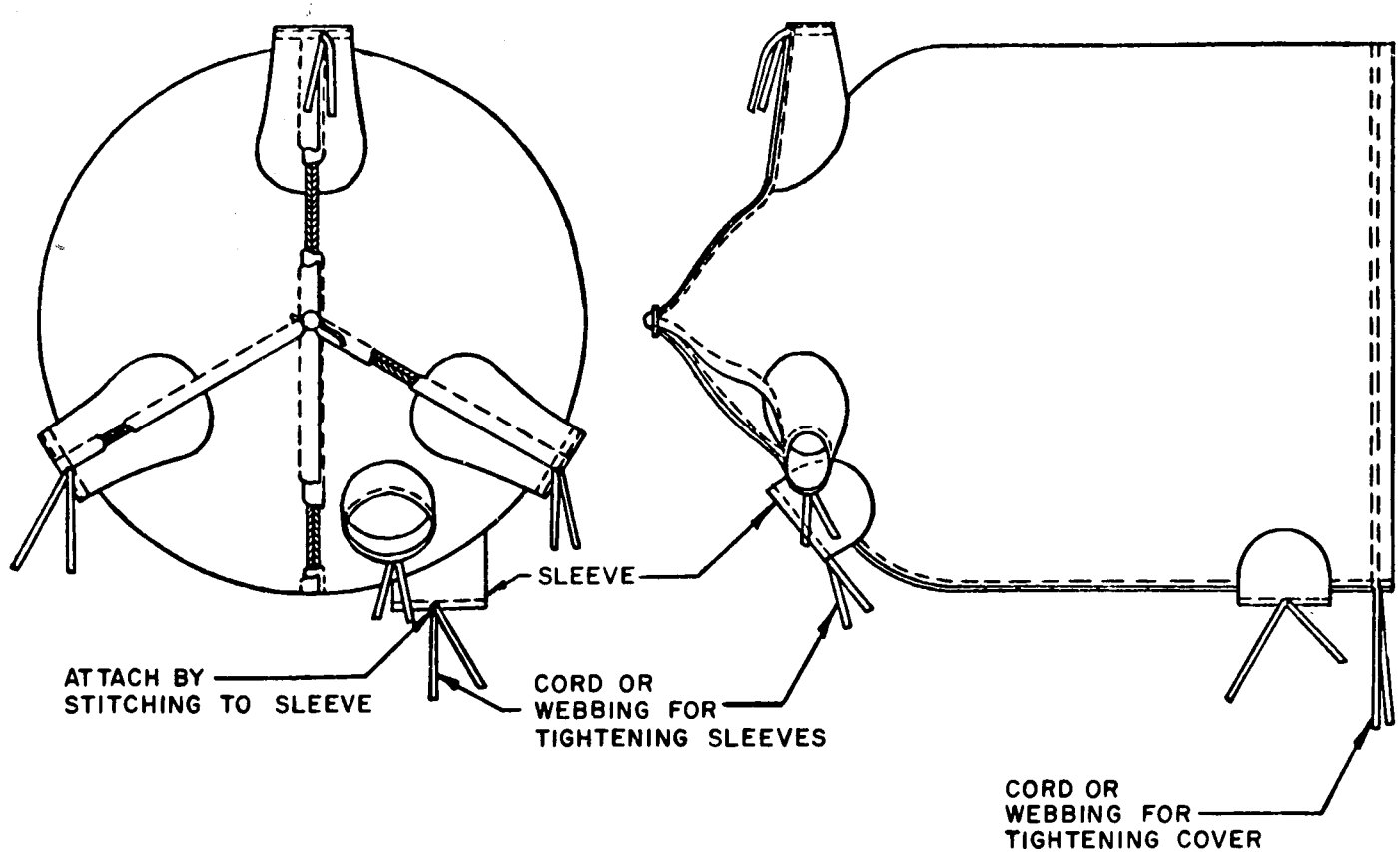


FIGURE 3. Typical Cover for Air-Cooled Engines (3-Blade Propeller)

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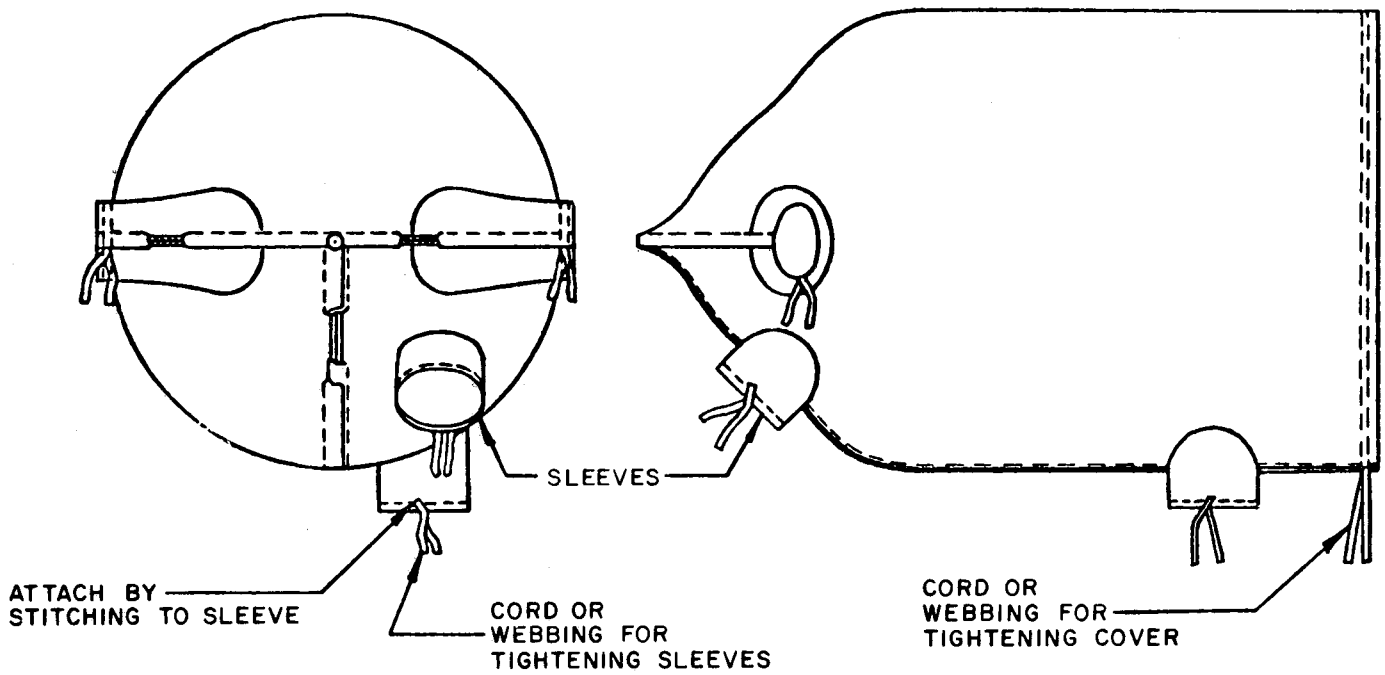


FIGURE 4. Typical Cover for Air-Cooled Engine (2-Blade Propeller)

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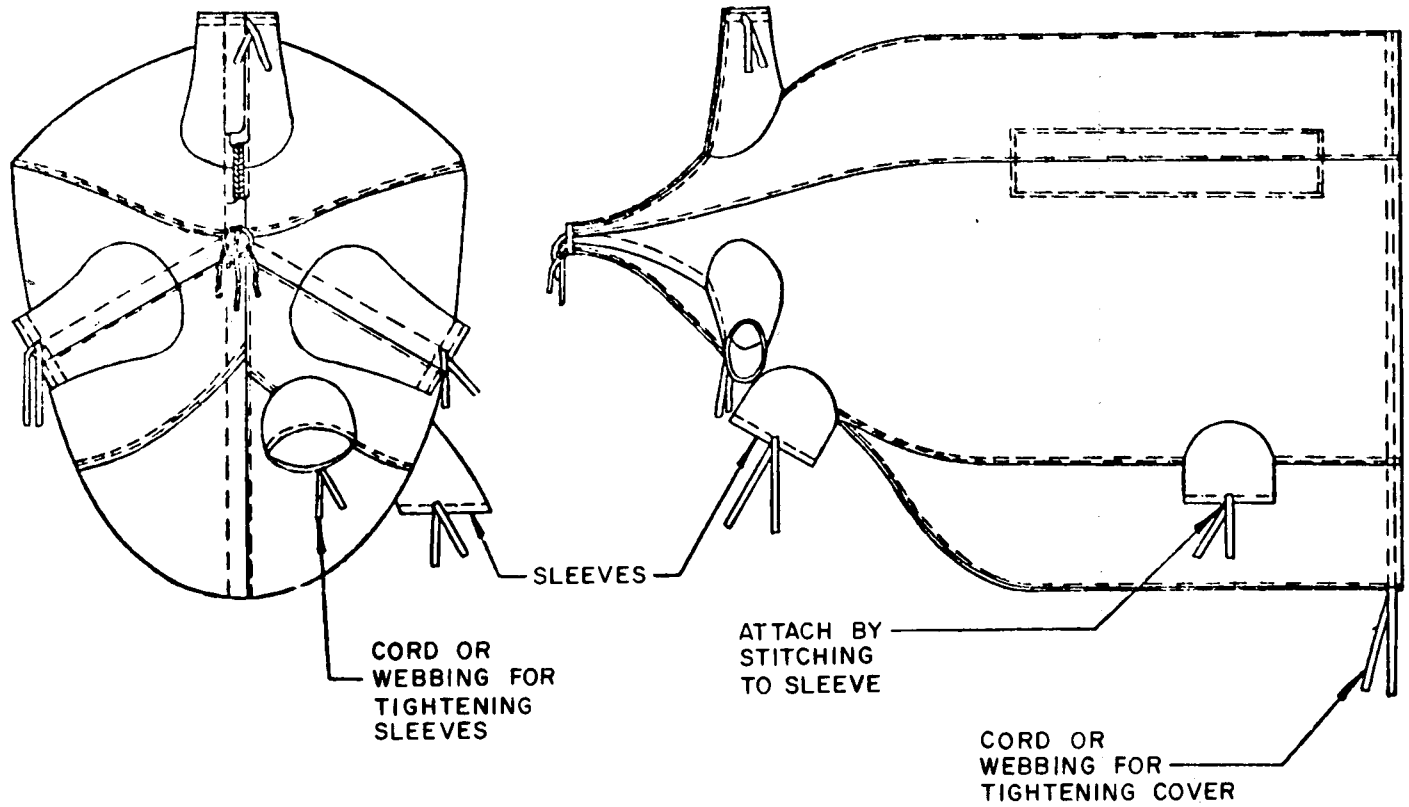


FIGURE 5. Typical Cover for Liquid-Cooled Engines (3-Blade Propeller)

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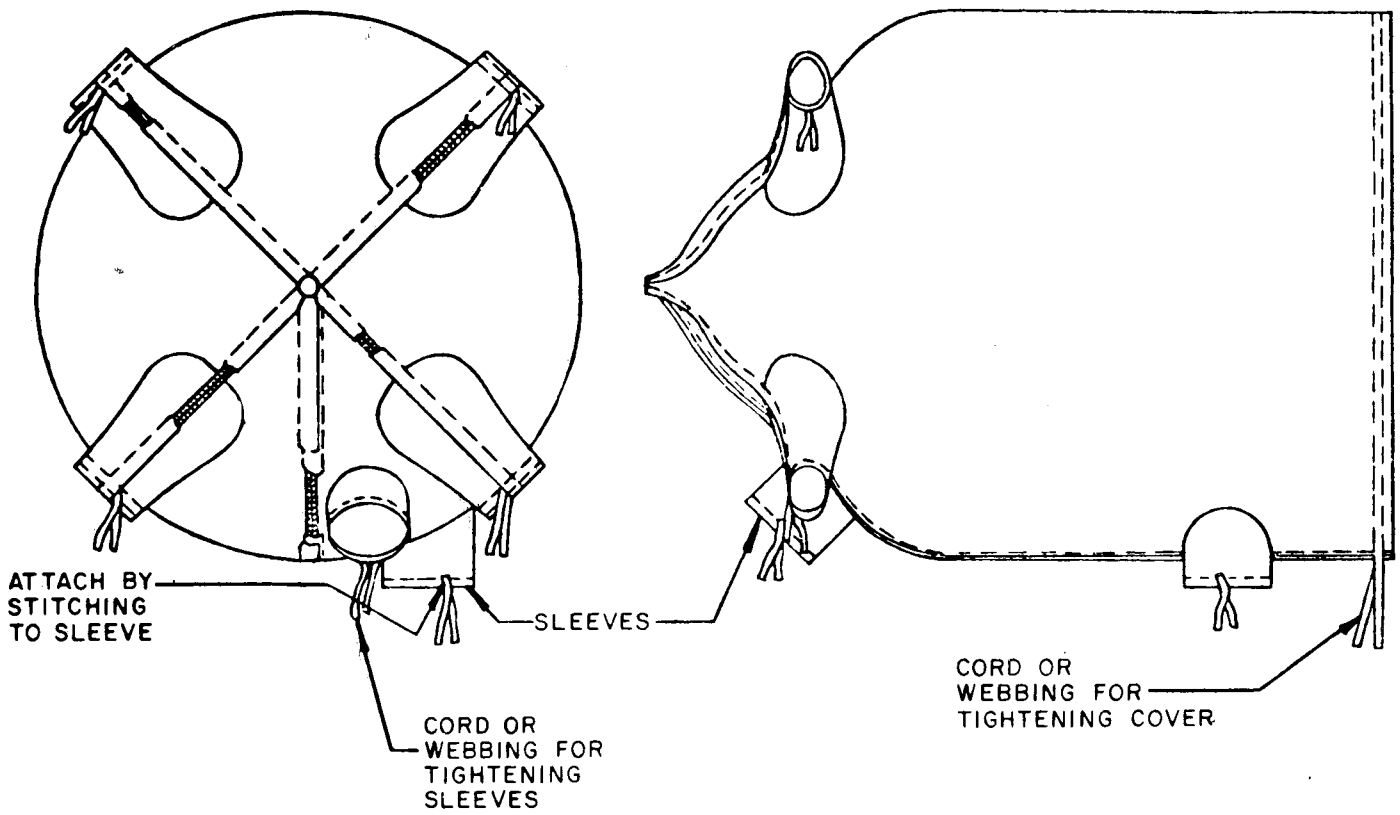


FIGURE 6. Typical Cover for Air-Cooled Engines (4-Blade Propeller)

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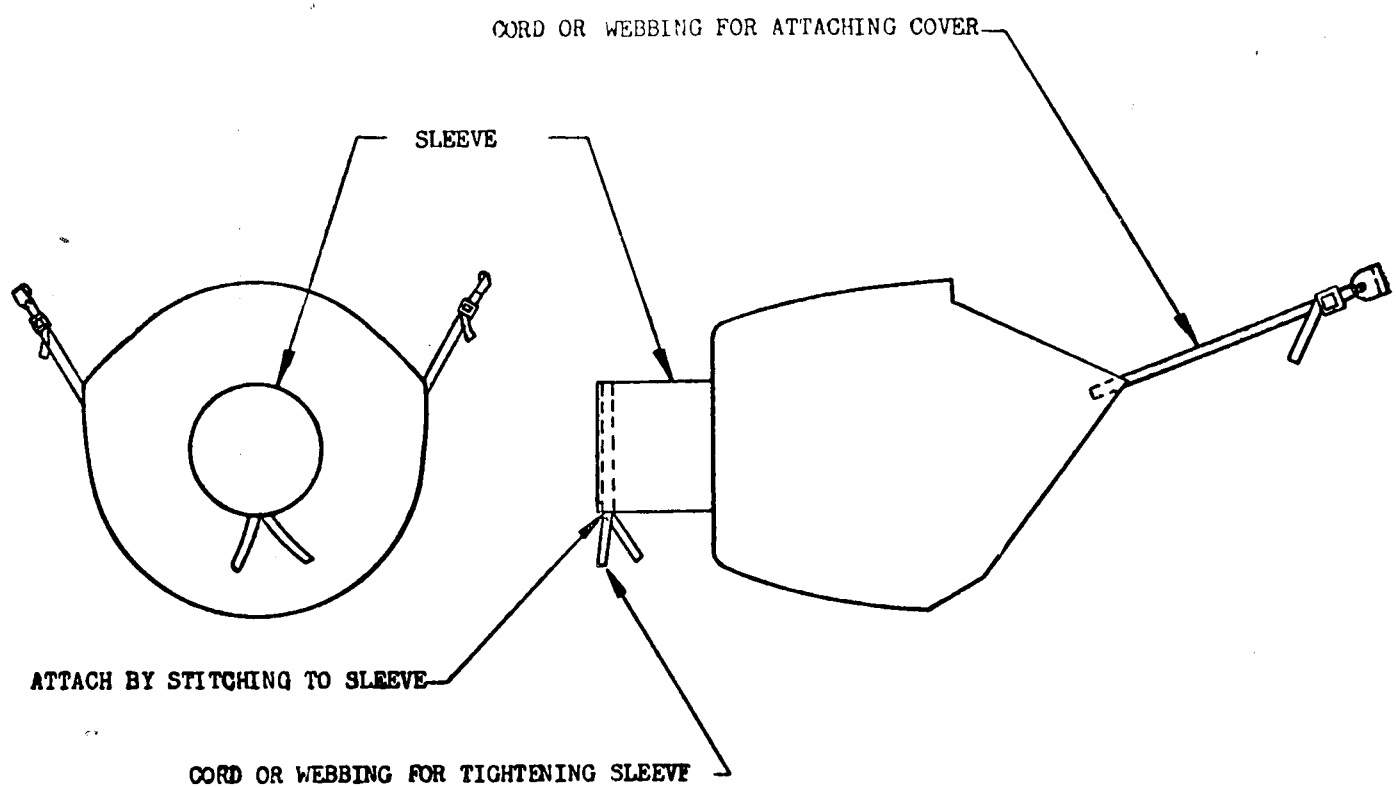


FIGURE 7. Typical Nose Cover for Jet Aircraft Engine.

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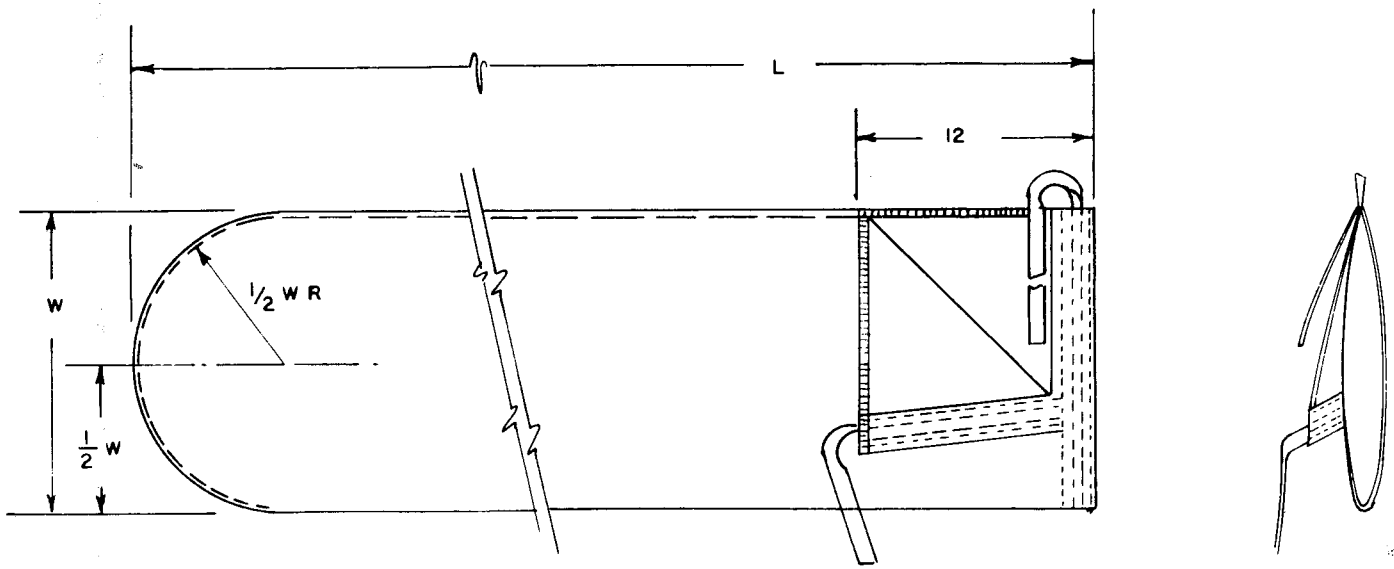
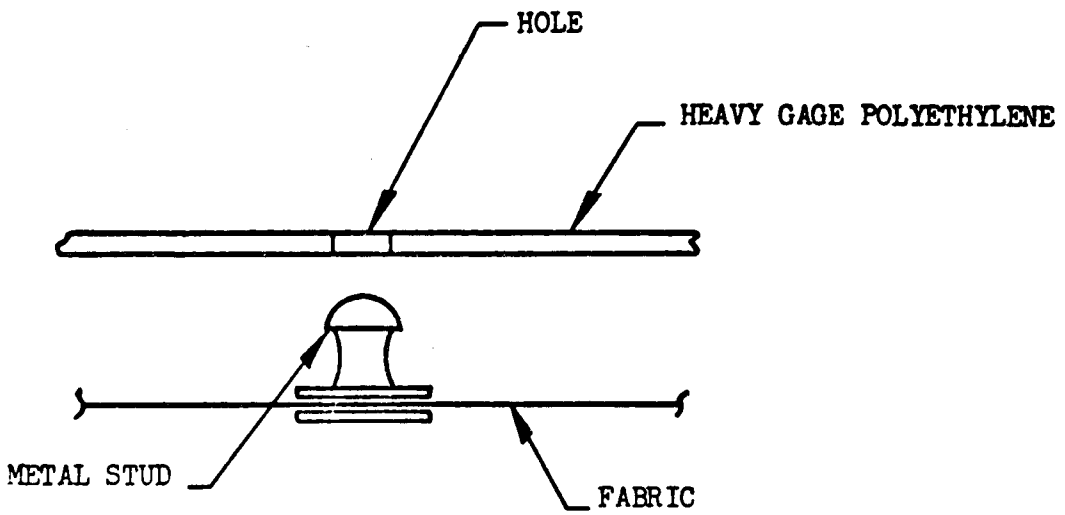
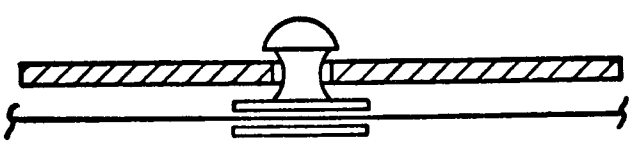


FIGURE 8. Typical Propeller and Rotor Blade Cover

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OPEN POSITION



CLOSED POSITION
CROSS SECTION

FIGURE 9. Snap Fastener (for type II only)

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   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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